

Figure 3.13 Interpolation of yellow to blue based on hue, saturation, and lightness (HSL) results in a different set of intermediary colors from the same two starting values.

You'd be forgiven if you expected the colors in figure 3.12 to range from yellow to green to blue. The problem is that the default interpolator in the scale we used is mixing the red, green, and blue channels numerically. We can change the interpolator in the scale by designating one specifically, for instance, using the HSL representation of color (figure 3.13) that we looked at earlier:

```
var ybRamp = d3.scale.linear()
  .interpolate(d3.interpolateHsl)
  .domain([0,maxValue]).range(["yellow", "blue"]);
```

Setting the interpolation method for a scale is necessary when we don't want it to use its default behavior, such as when we want to create a color scale with a method other than interpolating the RGB values.

D3 supports two other color interpolators, HCL (figure 3.14) and LAB (figure 3.15), which each deal in a different manner with the question of what colors are between blue and yellow. First, the HCL ramp:

```
var ybRamp = d3.scale.linear()
  .interpolate(d3.interpolateHcl)
  .domain([0,maxValue]).range(["yellow", "blue"]);
```

Finally, the LAB ramp:

```
var ybRamp = d3.scale.linear()
  .interpolate(d3.interpolateLab)
  .domain([0,maxValue]).range(["yellow", "blue"]);
```



Figure 3.14 Interpolation of color based on hue, chroma, and luminosity (HCL) provides a different set of intermediary colors between yellow and blue.

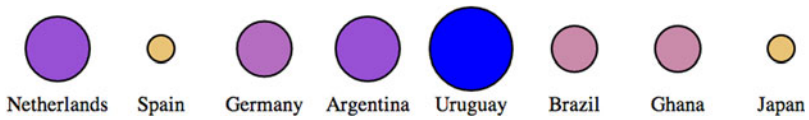


Figure 3.15 Interpolation of color based on lightness and color-opponent space (known as LAB; L stands for *lightness* and A-B stands for the color-opponent space) provides yet another set of intermediary colors between yellow and blue.